

Dinner Pool Report

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Introduction to COVID-19

The COVID-19 pandemic, caused by the SARS-CoV-2 virus, emerged in Wuhan, China, in December 2019 and quickly escalated into one of the most significant global health crises of modern times.

Transmitted primarily through respiratory droplets and contaminated surfaces, the virus spread rapidly across countries, prompting widespread illness, overwhelmed healthcare systems, and emergency public health responses.

Believed to have originated from animal-to-human transmission—likely involving bats and an intermediate host—COVID-19 has caused millions of deaths and left lasting health, social, and economic impacts worldwide

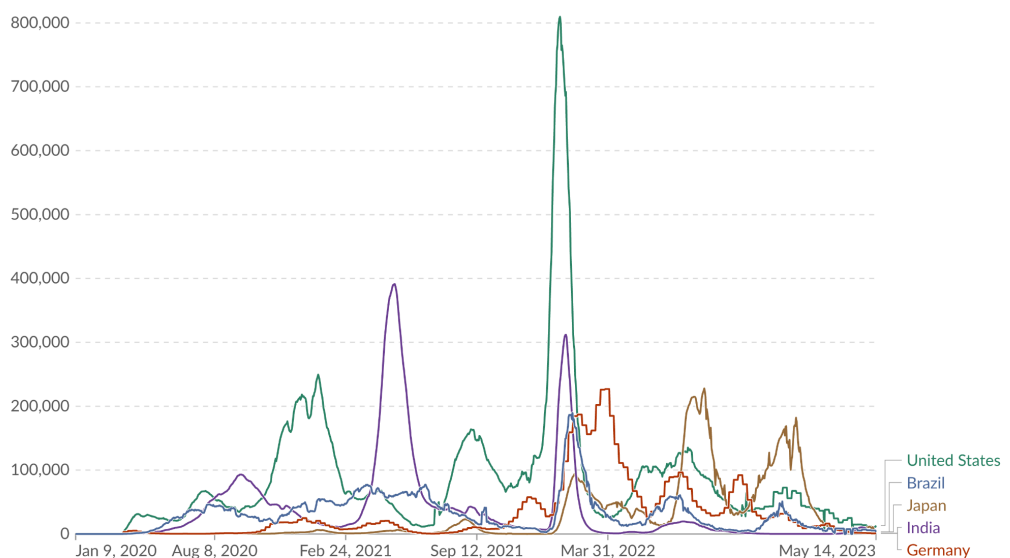
Overview of the Pandemic and Its Effects

COVID-19 had a global reach, with countries like USA, Italy, India and China being severely affected.

Its progress and effects were in waves, with the second wave being devastating in terms of cases and deaths.

Daily new confirmed COVID-19 cases

7-day rolling average. Due to limited testing, the number of confirmed cases is lower than the true number of infections.



Data source: World Health Organization (2025)

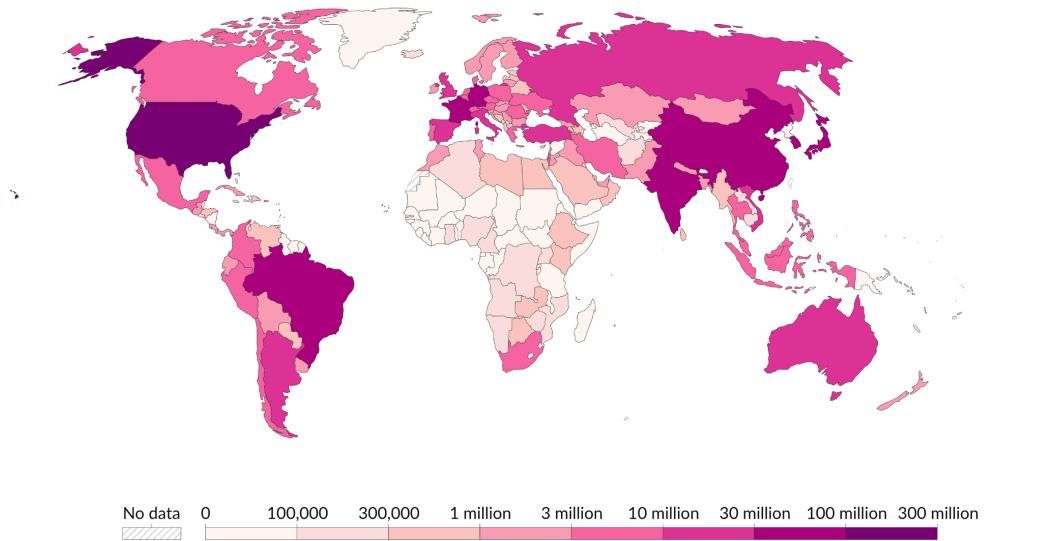
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Timeseries Plot of Daily New Confirmed Cases for USA, Brazil, Japan, India and Germany

Cumulative confirmed COVID-19 cases, Mar 2, 2025

Due to limited testing, the number of confirmed cases is lower than the true number of infections.

Our World
in Data



Data source: World Health Organization (2025)

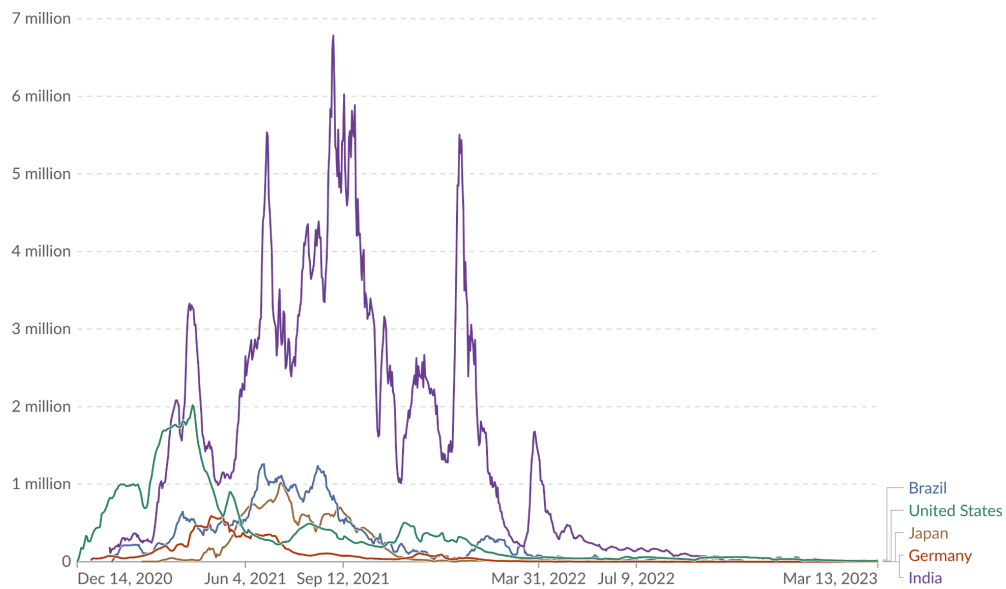
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Worldwide Map of Confirmed Cases as of March 2nd 2025

Daily number of people receiving a first COVID-19 vaccine dose

7-day rolling average

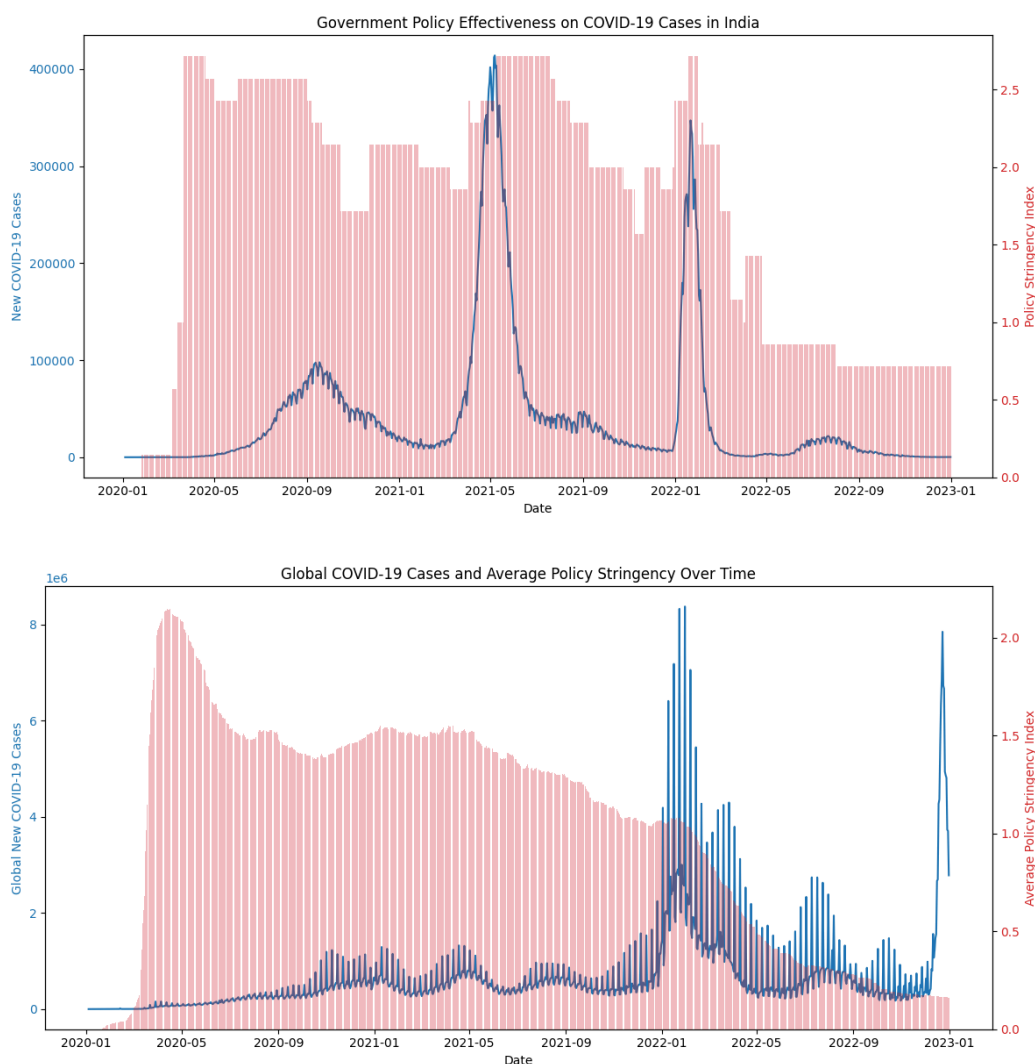
Our World
in Data



Data source: Official data collated by Our World in Data (2024); World Health Organisation (2025)

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Timeseries Plot of Daily Number of People Receiving First COVID Vaccine Dose for USA, Brazil, Japan, India and Germany



About Variants & Mutations

SARS-CoV-2, like all viruses, undergoes mutations over time. These genetic changes can lead to the emergence of new variants—some of which may spread more easily, cause more severe illness, reduce vaccine effectiveness, or affect diagnostic accuracy. Variants arise from different viral lineages and often share mutations that offer a survival advantage, allowing them to replace earlier strains.

To manage their impact, the World Health Organization (WHO) classifies variants into three categories:

- Variant Under Monitoring (VUM): Requires observation due to potential health risks.
- Variant of Interest (VOI): Carries genetic changes that may influence transmissibility or severity.
- Variant of Concern (VOC): Demonstrates significant impacts such as increased spread, immune evasion, or reduced treatment effectiveness. VOCs are assigned Greek letter names.

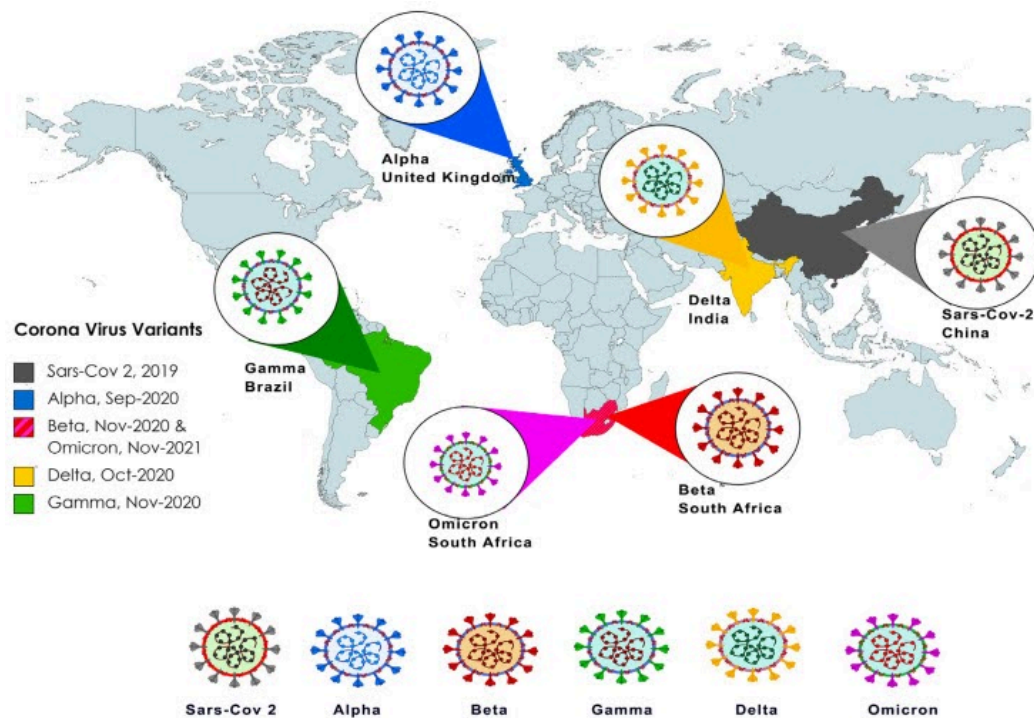
Key VOCs that shaped the pandemic include:

- Alpha (B.1.1.7) – More transmissible than the original strain.
- Beta & Gamma (B.1.351 & P.1) – Showed improved immune evasion.
- Delta (B.1.617.2) – Highly transmissible, linked to severe outbreaks.

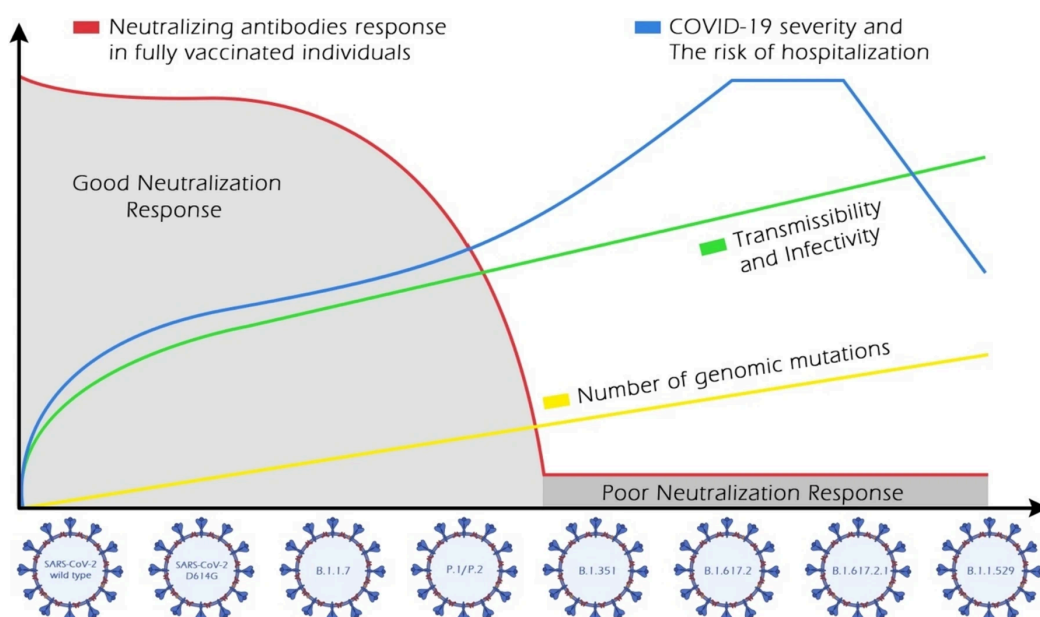
- Omicron (B.1.1.529) – Carried numerous mutations enabling immune escape, though with generally lower severity.

Genomic surveillance is crucial in tracking these variants, informing public health strategies, and guiding vaccine updates and outbreak control.

Map of Emergence of COVID-19 Variants



Risk Factors Comparison of Variants



Our Dashboard

Our project dashboard, titled "**The Pandemic: Trends in COVID-19 Variants**", was developed to offer a clear, data-driven perspective on the evolving landscape of COVID-19 variants. It serves as an analytical tool to understand trends, patterns, and implications of variant emergence and spread.

The dashboard is composed of multiple modules, each tailored to a specific analytical objective:

- **Prevalence Over Time** (Time Series of Variants)

This section displays the global time series data for COVID-19 cases per million, broken down by major variants—**Alpha, Beta, Gamma, Delta, and Omicron**. Each subplot illustrates the rise and fall of a specific variant over time.

Key Observations:

- **Alpha & Beta (Early 2021):** Emerged early in the pandemic and initially contributed to surges, but were quickly overtaken by more transmissible variants.
- **Gamma & Delta (Mid-2021):** Delta, in particular, dominated due to higher transmissibility and immune escape capabilities.
- **Omicron (Late 2021 - 2022):** Showed an explosive increase in cases globally, quickly surpassing all previous variants in dominance.

Purpose:

This visualization correlates variant dominance with major COVID-19 waves, providing a clear timeline of variant progression and highlighting the global public health impact of each wave.



- **Trend Analysis** (STL Decomposition)

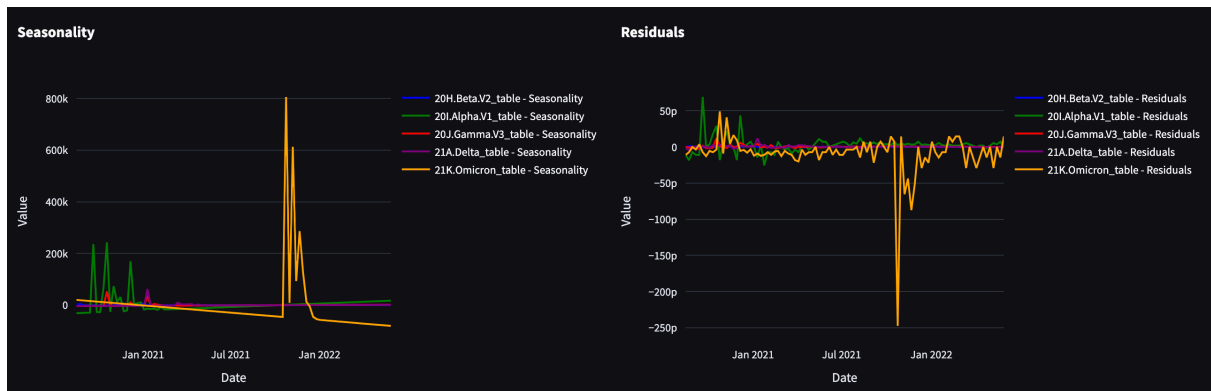
STL (Seasonal-Trend Decomposition using LOESS) breaks down time series data into three components: **Trend**, **Seasonality**, and **Residuals**, giving a deeper understanding of the dynamics behind variant spread.

Components:

- **Trend:** Indicates whether a variant is consistently increasing or decreasing over time.
- **Seasonality:** Identifies repeating patterns, often linked to climate, population behavior, or viral biology.
- **Residuals:** Captures random noise or unexplained spikes, possibly from new mutations or policy changes.

Why STL is Useful:

- It separates **short-term fluctuations from sustained trends**, improving clarity.
- It reveals **seasonal patterns**, helping anticipate future waves.
- It identifies **unexpected deviations**, signaling events like new variant emergence or underreporting.



• Seasonal Trends (Seasonal Emergence)

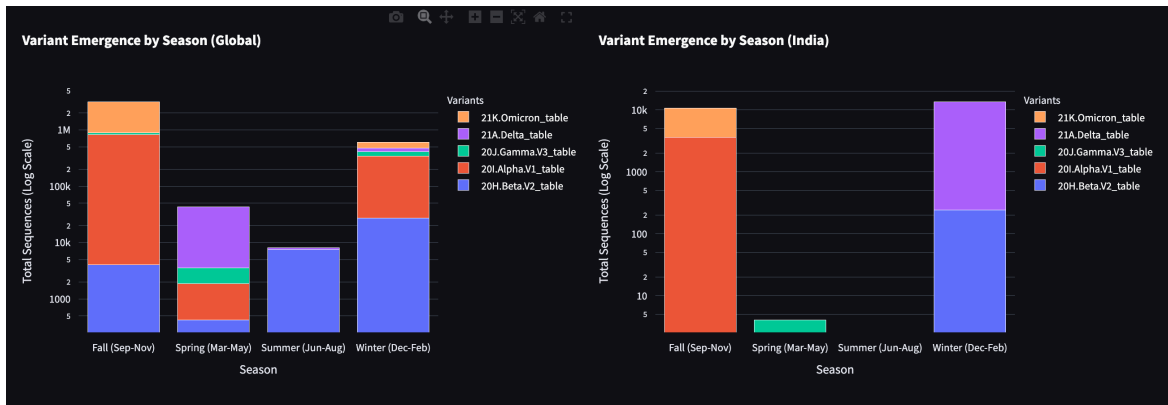
This section compares **seasonal emergence trends** of COVID-19 variants globally and specifically in India, helping highlight geographical differences.

Key Insights:

- **Delta:** Originated in India and surged earlier locally (Spring 2021) before becoming globally dominant later.
- **Omicron:** Peaked globally in Fall 2021, while India experienced a smaller surge.
- **Alpha:** Significant globally in Winter 2020-21, but India was already grappling with Delta.
- **Beta & Gamma:** Had a stronger presence globally, with limited impact in India.

Purpose:

This comparative analysis showcases how variant timing and impact vary across regions, influencing localized policy responses.



- **Forecasting Models** (ARIMA Forecasting of Cases)

This module allows users to forecast the future case trajectory of a selected COVID-19 variant using the **ARIMA (AutoRegressive Integrated Moving Average)** model.

How ARIMA Works:

- **AutoRegression (AR):** Predicts future values using past observations.
- **Integration (I):** Removes long-term trends to stabilize the data.
- **Moving Average (MA):** Smooths out forecast errors by averaging previous error terms.

Usage:

- Users can select a specific variant and a time frame (e.g., 6 weeks) to view predicted trends.
- Forecasts help estimate variant spread under the assumption that **past patterns continue**, although **unforeseen factors** like new mutations may impact accuracy.



- **Mutation Rates** (Mutation Frequency)

This section educates users on the relationship between **mutations and variant formation**. It features a heatmap that highlights **mutation frequency and dominance** across key variants.

Key Takeaways:

- **Mutation Profiles:** Each variant has distinct sets of dominant mutations (e.g., S:P681R in Delta, multiple mutations in Omicron).
- **Impact on Spread:** Certain mutations increase transmissibility or aid immune escape, making some variants more successful.
- **Evolution Over Time:** Variants replace one another as **more adaptive mutations** arise—Alpha → Delta → Omicron.
- **Convergent Evolution:** Some mutations recur across different variants due to their survival advantages.

Purpose:

This module helps visualize the **genetic evolution** of the virus and connects mutation trends with **epidemiological waves**.



- **Simulation of Spread** (SEIR³D Pandemic Simulator)

Our dashboard includes a sophisticated simulation tool based on the **SEIIRD** model, an extension of the classical SEIR model, tailored for COVID-19.

Compartments:

- **S (Susceptible)** – Uninfected individuals at risk.
- **E (Exposed)** – Infected but not yet infectious.
- **I_{asym} (Asymptomatic Infected)** – Can transmit without symptoms.

- **I_sym (Symptomatic Infected)** – Show symptoms but not critical.
- **I_sev (Severe Infected)** – Critically ill, possibly hospitalized.
- **R (Recovered)** – Recovered and immune.
- **D (Dead)** – Mortality count.

Simulation Options:

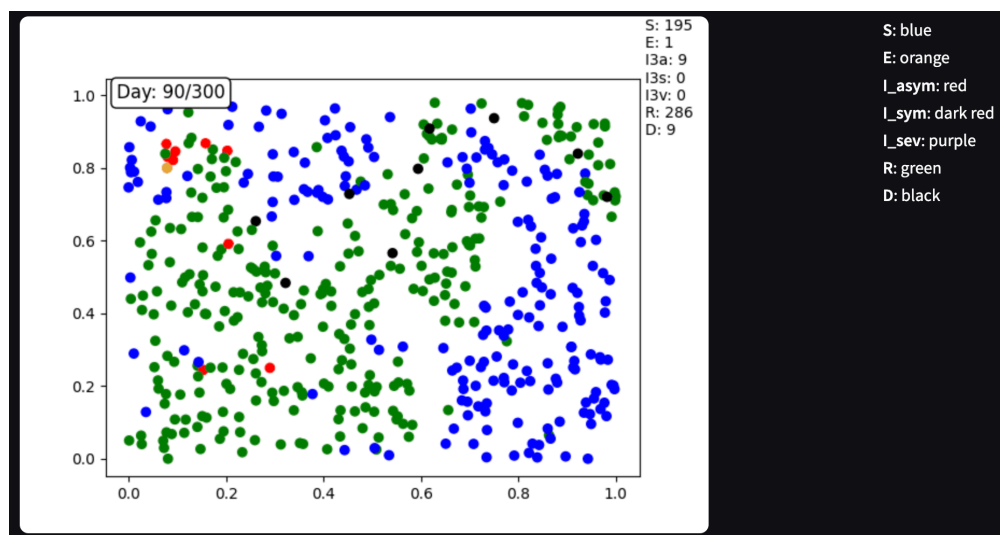
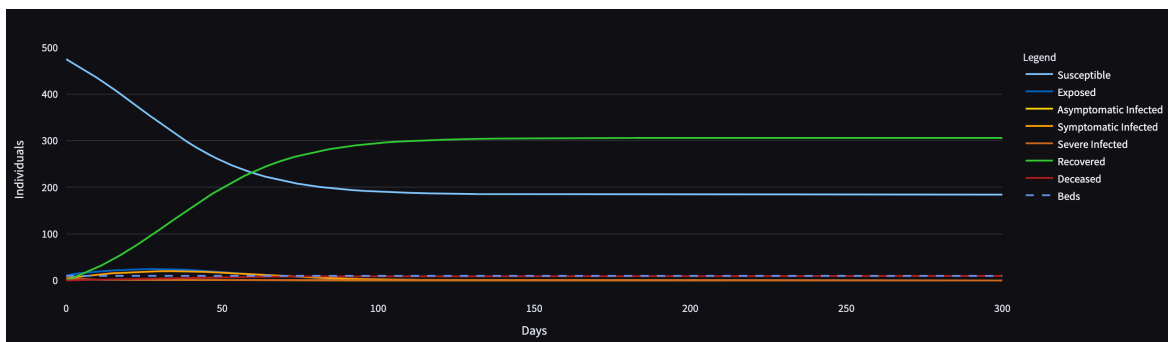
- **Variant-Based Simulation:** Uses default transmission parameters for known variants.
- **Parameter-Based Simulation:** Users can adjust key parameters like transmission rate, recovery rate, etc., for custom modeling.

Insights:

- Early pandemic variants were accurately simulated.
- Later, highly infectious variants like Omicron diverged from model predictions due to **complex real-world factors** like vaccination and policy interventions.

Purpose:

This tool models disease spread under different conditions, aiding in **policy evaluation**, **capacity building**, and **public health strategy testing**.



Main Dashboard Image:

THE PANDEMIC: TRENDS IN COVID-19 VARIANTS

Introduction

The COVID-19 pandemic was one of the most significant global health crises in modern history, caused by the SARS-CoV-2 virus. First identified in late 2019, the virus spread rapidly, leading to widespread illness, healthcare strain, and unprecedented public health responses.

Understanding COVID-19 Variants

Viruses constantly evolve, and COVID-19 variants emerged as the virus spread worldwide. Some of these changes had major public health implications, leading to the classification of variants into three categories:

- **Variant Under Monitoring (VUM)** – A variant that requires close tracking due to potential risks.
- **Variant of Interest (VOI)** – A variant with genetic changes that may affect its spread, severity, or treatment.
- **Variant of Concern (VOC)** – A variant that significantly impacts disease severity, vaccine effectiveness, or healthcare systems.

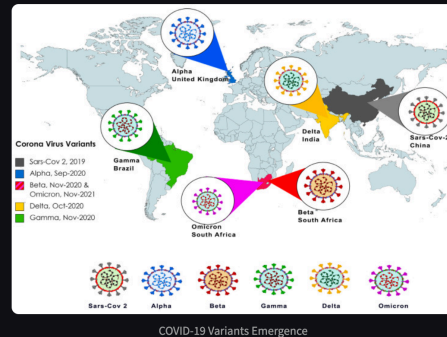
What This Dashboard Offers

This interactive dashboard provides a quick and clear overview of COVID-19 and its variants, tracking their prevalence, mutations, and seasonal patterns.

The variants we've focused on in this dashboard are:

- Alpha (B.1.1.7)
- Beta (B.1.351)
- Gamma (B.1.1.28.1)
- Delta (B.1.617.2)
- Omicron (B.1.1.529)

These variants played a key role in shaping the course of the pandemic, influencing public health responses and



These tools collectively provide insights for public health decision-making, policy responses, and academic research. The variants chosen shaped the course of the pandemic and were instrumental in influencing vaccine development strategies.

Dashboard Applications

The dashboard from this research can be useful in a number of fields such as :

- **Healthcare Resource Planning:**
Predict hospitalizations and resource needs (e.g., ICU beds, ventilators) over time for better management
- **Public Education and Communication:**
Serve as an interactive tool to explain COVID-19 dynamics to the public or students.
- **Research and Scenario Analysis:**
Allow researchers to test various scenarios by adjusting parameters such as transmission rates or immunity waning.

LimitationsPrevalence Over Time

Some current limitations of our dashboard are:

- **Limited Data Availability and Regional Bias**
Some countries contribute more genomic data than others, leading to potential underrepresentation of variants in low-resource settings and introducing geographic bias in trend analysis.
- **Inability to Predict Novel Variant Behavior**
While the dashboard can forecast trends based on known variants and mutations, it cannot anticipate the emergence or behavior of entirely new variants with unknown genetic profiles. This limits its predictive power in rapidly changing outbreak scenarios involving novel or recombinant strains.

Future Work

This project can be further extended to analyse and simulate the progress of COVID-19 and other infectious diseases in different situations :

- Extend to include different at-risk groups
- Use more external factors like vaccination rates, mobility of public and implementation of government policies (lockdown, closing of schools).

These directions can give us more comprehensive and realistic results.

The dashboard can also be further improved by including more information about global safety measures and related data.

Resources:

- <https://github.com/hodcroftlab/covariants/blob/master/web/public/data/perCountryDataCaseCounts.json>
- https://github.com/adyanshkakran/disaster_encouragement
- <https://www.ecdc.europa.eu/en/covid-19/variants-concern>
- https://en.wikipedia.org/wiki/Variants_of_SARS-CoV-2
- <https://covariants.org/cases>
- <https://cov-lineages.org/resources.html>
- [SARS-CoV-2 journey: from alpha variant to omicron and its sub-variants | Infection](#)
- [Alpha, Beta, Gamma, Delta, Omicron: COVID-19 Variants Explained](#)
- [From Alpha to Omicron: How Different Variants of Concern of the SARS-Coronavirus-2 Impacted the World - PMC](#)
- **[WHO COVID-19 dashboard](#)**
- **[GISAIID - hCoV-19 Variants Dashboard](#)**
- **[Nextstrain / ncov / gisaid / global / 6m](#)**
- **[COVID-19 - ETL](#)**
- **[Extensions of the SEIR model](#)**